

Humanoid Joint Drive

Research Overview & Test Plan

July 15, 2026

Summary

Designing and validating a 96V GaN-based motor drive (EPC91200, EPC2305 eGaN FETs) for humanoid robot leg joints. Compared against a 48V Si-equivalent baseline, this project aims to prove:

- **Higher efficiency** — lower phase current at doubled voltage reduces copper losses by up to $4\times$
- **Lower thermal load** — cooler FETs and windings at the same shaft power output
- **Reduced total drive loss** — switching, conduction, and iron losses compared across voltage and frequency
- **Improved backdrivability (Z-width)** — faster GaN switching enables smaller dead-times, reducing torque distortion and improving transparency
- **Effective regenerative braking** — zero reverse-recovery in GaN allows clean 4-quadrant operation; regen energy captured and quantified
- **Better gait tracking** — higher control bandwidth and lower friction enables more accurate trajectory following
- **More human-like motion** — passive policy (Duke + ECO hybrid) reduces unnecessary joint activation, lowering CoT and producing smoother, more natural gait

Phase	Description	Motor	Hardware
1 — Simulation	Virtual dyno baseline; FOC pre-tuning	Both modelled	MATLAB/Simulink
2a — Rig Validation	Validate test setup at lower power	AKE80-8 (ankle)	EPC91200
2b — Characterisation	Full test matrix — actual thesis data	Custom AKE90-KV35	EPC91200 (same)
3 — Leg Validation	Integrate into physical leg, test gait	Custom AKE90-KV35	EPC91200

Phase 1 — Simulation

Simulink/Simscape virtual dynamometer used to generate theoretical baselines and pre-tune FOC gains before any hardware runs. All simulation plots act as predicted reference curves that real data will be plotted against.

Test	Output
Torque $\times$ speed efficiency sweep (48V / 96V)	Efficiency contour maps (η vs T , ω) for both configs
Loss breakdown at each operating point	Stacked bar charts: P_{Cu} , P_{sw} (switching), P_{Fe} (iron)
Switching frequency sweep (20–100 kHz)	Total loss vs f_{sw} ; motor loss vs f_{sw} ; inverter loss vs f_{sw}
Voltage comparison at fixed operating point	η , I_{RMS} , loss split — 48V vs 96V side by side
FOC PI gain tuning	Pre-validated current loop gains for hardware

Phase 2 — Drive Characterisation Tests

Tests are run incrementally — **one variable changes at a time** — so each result directly attributes an improvement to a specific design choice. Phase 2a runs the AKE80-8 to validate the test rig; Phase 2b repeats with the AKE90-KV35 for thesis data.

Test Hardware Options

Path	Setup	What it enables
A — Active Dyno (preferred)	Motor coupled to a load motor with torque sensor	Full efficiency map, regen tests, any operating point
B — Pendulum Arm	Motor drives a calibrated pendulum arm	Discrete efficiency points, thermal, controller comparison

T1 — Voltage Comparison: 48V vs 96V

Fixed: same motor, same GaN board, standard RL controller, same f_{sw} . Variable: Bus voltage.

Measured	Expected Result
$\eta(T, \omega)$ efficiency map	96V map shows higher η across most of the grid
I_{RMS} at same torque output	$\sim 50\%$ lower at 96V
Copper loss $P_{Cu} = 3 \cdot I^2 \cdot R$	$\sim 75\%$ lower at 96V
Switching loss P_{sw}	$\sim 2\times$ higher at 96V (same f_{sw} , higher voltage per switch)
Peak FET and winding temperature	Lower at 96V for same shaft power

T2 — Switching Frequency Sweep

Fixed: 96V bus, standard RL controller. Variable: f_{sw} swept at 20/40/60/80/100 kHz. *Also run at 48V to show comparison.*

Measured	48V Result	96V Result
Total actuator loss vs f_{sw}	Increases with f_{sw} (Si-like)	Flatter curve (GaN switching loss scales less severely)
Inverter switching loss vs f_{sw}	Linear increase	Linear increase, but from a lower base at same duty
Phase current ripple ΔI_{pp} vs f_{sw}	Decreases as $\sim 1/f_{sw}$	Same

T3 — Dead-Time Sweep

Fixed: 96V bus, optimal f_{sw} , standard RL. Variable: dead time swept from 500ns down to 50ns.

- **Measured:** Phase current THD, torque ripple, backdrive drag torque (Z-width) at $I_{cmd} = 0$.
- **Expected:** All decrease linearly as dead time reduces, until shoot-through boundary is reached.

T4 — Regenerative Braking (Path A only)

Fixed: 96V bus, optimal f_{sw} , optimal DT. Variable: braking torque level.

- **Measured:** Regen energy captured ($E_{regen} = \int V_{bus} \cdot |I_{bus}| dt$), V_{DS} switching waveforms, bus voltage transients.
- **Expected:** Demonstrates zero Q_{rr} commutation vs Si body diode ringing.

T5 — Controller Comparison

Fixed: 96V, optimal f_{sw} , optimal DT. Variable: control law. Each tested in sequence.

Controller	Architecture Summary
Standard RL	Fixed PD torque controller with soft energy penalty in reward. Known to collapse at high penalty weight.
Duke Humanoid	Policy outputs joint targets + per-joint activation $\alpha \in [0, 1]$. Torque gated as $\tau = \alpha \cdot PD(q^*, q)$. Passive reward $1/\ \alpha\ $ encourages joint relaxation.
ECO	Energy treated as hard inequality constraint via PPO-Lagrangian. Lagrange multiplier self-tunes — prevents policy collapse.
Modified (Ours)	Combines Duke’s α gating with ECO’s constrained training. Adds regen bonus tied to gait phases (heel strike, swing deceleration) to exploit high voltage GaN Quadrant IV operation.

Metrics (all controllers): Cycle-averaged efficiency η , Electrical Cost of Transport (CoT) per stride, I_{RMS} , Peak FET temperature, Gait tracking RMS error, Step response (bandwidth).

Phase 3 — Leg Build & System Validation

The validated drive from Phase 2 is integrated into a physical leg to validate the full system under real gait loading.

Joint	Motor	Notes
Hip pitch	Custom AKE90-KV35	Same frame + gear ratio as standard AKE90; re-wound for 96V
Knee	Custom AKE90-KV35	Same
Ankle	AKE80-8 (or custom)	8:1 planetary

Test	Data Collected
No-load joint sweep	Friction map, backlash, encoder calibration
Automated gait replay	Simulated trajectory replayed on hardware
Efficiency (sim vs real)	Measured η overlaid on Phase 1 simulation maps
Thermal soak	ΔT_{FET} and $\Delta T_{winding}$ vs time under repeated gait cycles
Controller comparison	Repeat T5 metrics on real leg — CoT, η , tracking quality

Parts & Requirements

Phase 2 — Path B (Pendulum — can start immediately)

Item	Qty	Source / Details
AKE80-8 motor & EPC91200 board	1	Available
Motor mount bracket	1	3D print PAHT-CF (brass inserts at bolt locations)
Table clamp / L-bracket	1	Buy / weld
Shaft coupler	1	Buy
Steel flat bar pendulum arm	1	Buy
Split-collar masses (0.5 kg + 1.0 kg)	2–3	Buy
Instrumentation	1	Lab (DAQ, 100V PSU, Oscilloscope, 3× Hall probes)

Phase 2 — Path A (Dyno — additional items)

Item	Qty	Source / Details
Load motor & driver (≥ 1.5 kW, regen)	1	Buy / borrow / lab?
Inline torque sensor (≥ 15 Nm)	1	Buy
Motor alignment & Dyno base plates	1	CNC aluminium

Phase 3 — Leg Build

Item	Qty	Source / Details
Custom AKE90-KV35 motors	2+	Custom wound
EPC91200 inverter boards	2-3	Additional units
Leg structural links	1 set	PAHT-CF (light load) or CNC aluminium (high load)
Encoders & Wiring harness	1 set	Buy / Make

Open Questions

1. **Load motor availability** — Is a load motor available in the lab? (Determines Path A vs B order)
2. **CNC access** — Lab or outsource? (Affects timeline for Path A dyno setup)
3. **Phase 3 leg spec** — Determines PAHT-CF vs aluminium